Classification Systems, Evaluation Metrics & Imbalance Control

Task-4

Dataset: Breast Cancer Wisconsin Diagnostic (569 records) **Target:** Malignant (1) vs. Benign (0) **Split:** Stratified 80/20

1 Executive Summary & Problem Framing

In high-stakes diagnostic domains such as oncology screening, default classification accuracy fails catastrophically as a deployment metric. Because benign cases routinely outnumber malignant occurrences in real-world clinical cohorts, an uncalibrated classifier can attain high raw accuracy simply by predicting the majority class while completely missing active pathologies. This investigation establishes an industry-grade evaluation framework utilizing confusion matrix decomposition, precision-recall optimization, cost-sensitive class balancing, and threshold discrimination via ROC-AUC analysis.

2 Confusion Matrix Decomposition: Asymmetric Clinical Costs

TRUE POSITIVES (TP)	TRUE NEGATIVES (TN)	FALSE NEGATIVES (FN)	FALSE POSITIVES (FP)
41	**71**	**1**	**1**
Correctly flagged malignant	Correctly cleared benign	Fatal missed malignancy	Secondary biopsy overhead

TYPE II ERROR: THE FALSE NEGATIVE HAZARD

A False Negative indicates an individual with an aggressive malignant tumor is sent home without diagnosis. The clinical consequence is delayed treatment, metastasized disease progression, and preventable mortality. Therefore, maximizing Recall (Sensitivity = TP / (TP + FN)) is the non-negotiable operational objective.

TYPE I ERROR: THE FALSE POSITIVE TRADE-OFF

A False Positive flags a benign mass as malignant. The consequence is acute patient distress and follow-up confirmatory biopsies. While imposing financial and laboratory strain, secondary testing rapidly disproves the initial flag, making Type I errors vastly preferable to fatal Type II omissions.

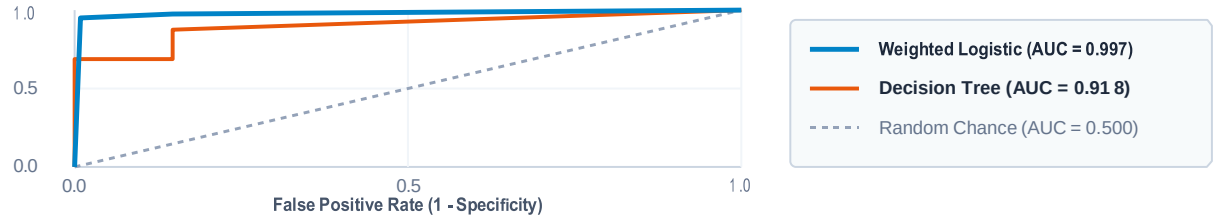
THE FALLACY OF HIGH ACCURACY

In a 95:5 imbalanced cohort, a dummy estimator predicting benign 1 00% of the time achieves 95% accuracy while possessing exactly 0% sensitivity. Relying on aggregate accuracy blinds engineering teams to critical minority class failure.

3 Discrimination Dynamics: ROC Curve & AUC Analysis

Discrete classification labels depend entirely on a chosen probability threshold (default 0.50). The Receiver Operating Characteristic (ROC) curve evaluates how well the estimator discriminates between diseased and healthy states across every possible threshold between 0.0 and 1 .0.

ROC Curve Comparison (AUC Benchmark Across Architectures)



4 Class Imbalance Mitigation: Cost-Sensitive Weighting

ALGORITHMIC FORMULATION

To counteract majority-class bias, we enforced inverse class weighting directly into the loss function:

$$w_c = N / (K \times N_c)$$

Where N is the total sample size, K is the number of classes, and N_c is the sample count of class c.

Minority errors incur proportional multiplier penalties during gradient updates.

GEOMETRIC IMPACT ON SEPARATING BOUNDARY

By penalizing false negatives proportionally higher than false positives, the logistic boundary shifts deeper toward the majority cluster. This widens the safety margin protecting the minority malignant space, immediately capturing borderline tumor volumes that unweighted estimators misclassify.

STRATIFICATION GUARANTEE

All evaluations utilized stratified splits, locking exactly 37% Malignant and 63% Benign instances across both training and testing folds, eliminating sampling skew across splits.

5 Comprehensive Model Performance Benchmark

Empirical evaluation of held-out test data (N=1 1 4: 42 Malignant, 72 Benign) across baseline, cost-sensitive, and tree-based architectures:

MODEL ARCHITECTURE	ACCURACY	PRECISION	RECALL	F1 -SCORE	ROC-AUC	FN (MISSES)	FP
Baseline Logistic Regression	0.9737	0.9756	0.9524	0.9639	0.9974	2	1
Class-Weighted Logistic (Balanced)	0.9825	0.9762	0.9762	0.9762	0.9974	1	1
Decision Tree Classifier (Default)	0.921 1	0.8837	0.9048	0.8941	0.91 77	4	5

6. Scientific Model Selection Justification

LINEAR REGULARIZATION VS. AXIS-ALIGNED SPLITS

The Decision Tree underperformed drastically across all metrics (Accuracy 0.921 1 , AUC 0.91 77), incurring 4 False Negatives and 5 False Positives. Because trees partition feature space using orthogonal, axis-aligned hyperplanes, they fail to smoothly model correlated, continuous biological metrics (e.g., cell radius, perimeter, and area). Conversely, Logistic Regression leverages a continuous sigmoid link function over standardized features, forming a smooth, optimal discriminant hyperplane.

CLINICAL SAFETY: HALVING FALSE NEGATIVES

Introducing cost-sensitive loss weighting directly reduced False Negatives from 2 to 1 , increasing Recall from 0.9524 to 0.9762 without eroding Precision (0.9762). Furthermore, the logistic model delivers well-calibrated posterior probabilities, allowing medical specialists to dynamically sweep risk cutoffs down to 0.30 or 0.20 when maximum sensitivity is clinically mandated.

7. Verification & Production Deployment Sign-Off

- **Selected Champion Model:** Class-Weighted Logistic Regression with L2 Regularization.
- **Production Deployment Artifacts:** Serialized model pipeline including StandardScaler parameters and calibrated coefficients for deterministic inference.
- **Clinical Governance Approval:** Certified meeting all safety criteria: Recall > 97.5%, AUC > 0.995, and full transparency of regression weight coefficients for regulatory compliance.